Climate change increases extreme weather and flood risks, and insurance companies are abandoning properties as risks increase. And as climate change worsens, flood risks are only projected to increase. “Because flooding is the primary vector of economic damages inflicted on local communities as demonstrated by the 2016-2019 hurricane seasons, and given the projected increase in destructive flooding as a result of climate change, there's an enormous need to more efficiently distribute financial risk due to climate change.”
https://www.kaggle.com/datasets/lynma01/femas-national-flood-insurance-policy-database

My approach was to use the OpenFEMA dataset of redacted NFIP policy transactions to build a supervised learning model to predict policy costs. The dataset had over 70 million records, so I sampled around ~350,000 records for training and ~350,000 records for testing. It had over 80 columns, many of which were missing many values, so I only kept latitude, longitude, and all columns with fewer missing values than those. With latitude and longitude, I projected the records on a map and found 2 outliers (Colombia, UK) I could eliminate. Other outliers I removed include dates before the earliest valid datetime[ns].

To prepare the data for modeling, I explored the boolean, categorical, and numerical features. I discovered that many of the boolean and categorical columns were extremely imbalanced; some boolean columns had well over 90% records with the same value. To avoid the curse of dimensionality, I kept the most balanced features: elevatedBuildingIndicator, ratedFloodZone (I got dummies for X and AE, with all the rest of the ratings effectively othered), postFIRMConstructionIndicator, primaryResidenceIndicator, and mandatoryPurchaseFlag. The numerical features needed transforming: latitude and longitude were cyclical, and many (especially the financial data) were significantly skewed. To do this, I projected latitude and longitude onto 3D Cartesian coordinates, reflected and log transformed left skewed data, and log transformed right skewed data. Finally, the data needed to be standard scaled for modeling. Because policy cost is simply the sum of the columns totalInsurancePremiumOfThePolicy, reserveFundAssessment, federalPolicyFee, and hfiaaSurcharge, I had to drop those columns to avoid multicollinearity.

Besides the baseline mean and median, I used linear regression, Ridge regression, Lasso regression, XGBoost, random forests, and neural nets. Random forests had the best performance upon cross-validation, so that ended up being my final model, with the most important feature being totalBuildingInsuranceCoverage. Unfortunately, it was not as predictive as I had hoped, as the R-squared was only around 0.5. As such, I would recommend calculating policy cost by summing its components, rather than predicting policy cost from the columns I chose. To build a better model, I would recommend incorporating the columns and rows I didn’t use due to time and computational constraints, while keeping in mind other limitations (e.g. missing values, curse of dimensionality).

For further research, I’d like to see how policy costs and other financial measures have changed with time. I would expect them to increase with time due to climate change and inflation among other factors; I would then like to explore how much each factor (e.g. climate change, inflation) explains the increase.